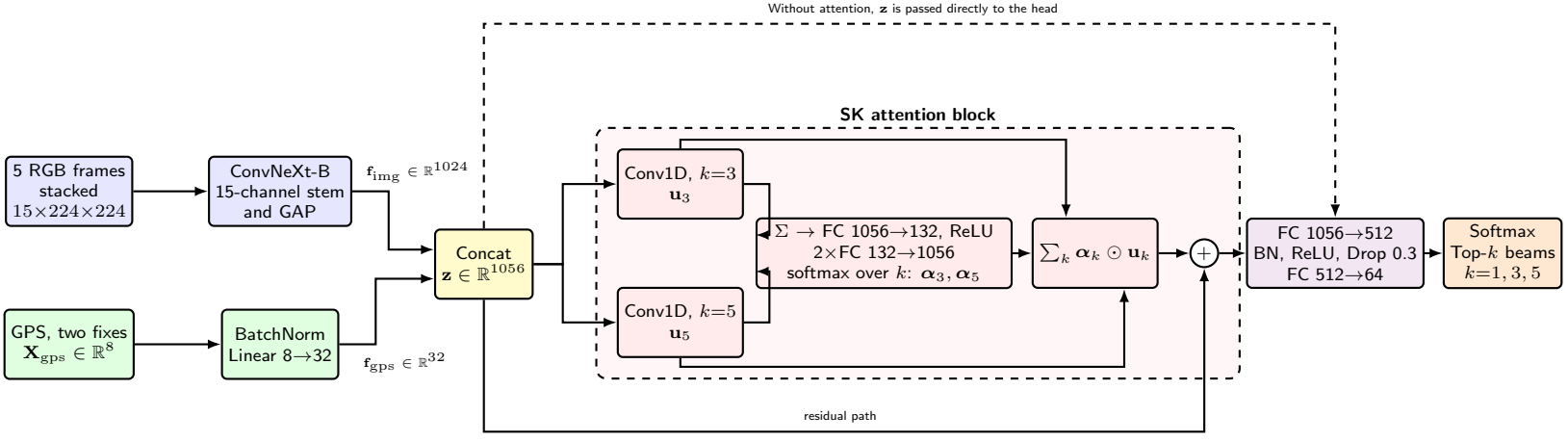

Without attention, z is passed directly to the head
SK attention block
5 RGB frames stacked 15×224×224
ConvNeXt-B 15-channel stem and GAP
$\mathbf{f}_{\text{img}} \in \mathbb{R}^{1024}$
GPS, two fixes $\mathbf{X}_{\text{gps}} \in \mathbb{R}^{8}$
BatchNorm Linear 8→32
$\mathbf{f}_{\text{gps}} \in \mathbb{R}^{32}$
Concat $\mathbf{z} \in \mathbb{R}^{1056}$
Conv1D, k=3 $\mathbf{u}_3$
Conv1D, k=5 $\mathbf{u}_5$
$\Sigma \rightarrow$ FC 1056→132, ReLU
2×FC 132→1056
softmax over k: $\boldsymbol{\alpha}_3, \boldsymbol{\alpha}_5$
$\sum_k \boldsymbol{\alpha}_k \odot \mathbf{u}_k$
FC 1056→512
BN, ReLU, Drop 0.3
FC 512→64
Softmax
Top-k beams
k=1, 3, 5
residual path